WEB-BASED ML MODEL EVALUATOR

System Validation Report

Project	Web-Based ML Model Evaluator (Team SE-Jackfruit)
Backlog item	BL-14 - Unit Test Report and System Validation Report
Validated against	Software Requirements Specification v1.0 (MLE-F-001 to 043, MLE-NF-001 to 005, MLE-SR-001 to 009)
Test run	4 November 2026 (automated suite 20:50 IST, manual cases the same evening)
Source of results	docs/test-results (pytest-verbose.txt, junit.xml, perf-100mb-output.txt) the Test Cases document and the Software Test Plan

1. Purpose

Validation answers the question: *does the finished system do what the requirements say?* This report checks every functional and non-functional requirement of the SRS against the acceptance criteria written there, using the test cases of the test plan. It then states for each requirement whether it is validated.

2. Method and environment

Item	Detail
System tested	The application on the main branch after the BL-11 merge (Flask web application, SQLite storage, scikit-learn evaluation).
Automated cases	pytest tests that drive the real application over HTTP with real model and CSV files, plus service-level tests. Command: python -m pytest tests -v -p no:cacheprovider --junitxml=junit.xml
Manual cases	NF_PF_03 (timing with a model close to 100 MB) and NF_US_01 (new-user usability check).
Environment	Windows, Python 3.14.6, pytest 9.1.1, run from the backend folder on 4 November 2026 at 20:50 IST
Pass rule	A requirement is validated when every test case linked to it passes and no acceptance criterion in the SRS is left uncovered.

3. Overall result

Test cases in the plan	Passed	Failed	Not executed	Defects found
86	86	0	0	0

All 86 cases of the plan were executed and passed. The automated suite ran 219 tests with no failures. Every requirement below is validated except MLE-SR-009 (TLS), which is a deployment check with no test; a few others carry a limitation noted in section 6.

4. Requirement-by-requirement validation

For each requirement: what the SRS asks, the test cases that check it, and the verdict. Case IDs are explained in the test plan; each case names its automated test or manual procedure.

Req.	Requirement (summary)	Test cases	Cases	Result	Verdict
MLE-F-001	Analyst registration. The system shall let a visitor register as an Analyst with a name, a valid email, a password of at least 8 characters and a matching confirmation, and shall reject blank fields, invalid emails, short passwords and an email that is already registered (ignoring letter case).	01_UT_01, 01_UT_02, 01_UT_03, 01_UT_04, 01_ST_01	5	5 of 5 pass	**Validated**
MLE-F-002	Login, session and logout. The system shall authenticate users by email and password, open the dashboard that matches their role, require a signed-in user for every page and API call except login and registration, and end the session on logout.	02_UT_01, 02_UT_02, 02_IT_01, 02_IT_02, 02_ST_01	5	5 of 5 pass	**Validated**
MLE-F-003	Admin accounts only through an Admin. The system shall never create an Admin through public registration. An Admin account shall be created only by an existing Admin, and the first Admin only from the command line.	SEC_02, 13_ST_01	2	2 of 2 pass	**Validated**
MLE-F-010	Dataset upload and validation. The system shall accept a CSV dataset of up to 50,000 rows, require the Analyst to name the target column when uploading, and reject non-CSV, corrupt, empty or oversized files and a missing or unknown target column, each with a reason.	03_UT_01, 03_UT_02, 03_UT_03, 03_UT_04, 03_IT_01, 03_IT_02	6	6 of 6 pass	**Validated**
MLE-F-011	Model upload and validation. The system shall accept trained model files (.pkl, .joblib) of up to 100 MB, and shall reject other file types, empty files, files over the limit and files that cannot be loaded as a model, each with a reason.	04_UT_01, 04_UT_02, 04_UT_03, 04_IT_01	4	4 of 4 pass	**Validated**
MLE-F-012	Model and dataset compatibility check. The system shall compare the input features a model expects with the columns of the dataset, show the result for each model and dataset pair, and shall not allow an evaluation of a model that is not compatible.	04_IT_02, 04_IT_03, 06_IT_02	3	3 of 3 pass	**Validated**
MLE-F-020	Task type selection and check. The system shall let the Analyst choose classification or regression and shall refuse to evaluate when the chosen task does not match the model type or the dataset target, giving the reason.	05_UT_01, 05_IT_01, 05_ST_01	3	3 of 3 pass	**Validated**
MLE-F-021	Classification metrics. The system shall run the model on the dataset for a classification task and calculate accuracy, precision, recall, F1-score (weighted for multiclass) and ROC-AUC. ROC-AUC shall be skipped, with a note, when the model gives no probabilities.	06_UT_01, 06_UT_03, 06_ST_01	3	3 of 3 pass	**Validated**

Req.	Requirement (summary)	Test cases	Cases	Result	Verdict
MLE-F-022	Regression metrics. The system shall run the model on the dataset for a regression task and calculate MAE, MSE, RMSE and R2, and shall not show classification metrics for a regression result.	06_UT_02, 06_ST_02, 07_UT_01	3	3 of 3 pass	**Validated**
MLE-F-023	Failed evaluation leaves nothing behind. The system shall stop an evaluation that fails during prediction, metric calculation or chart preparation, store no partial result and tell the Analyst why.	06_IT_02, 06_ST_03	2	2 of 2 pass	**Validated**
MLE-F-030	View metrics. The system shall show the stored metric values of a completed evaluation, and a clear message when the evaluation does not exist.	07_UT_01, 07_UT_02, 07_IT_01	3	3 of 3 pass	**Validated**
MLE-F-031	Visualizations. The system shall draw a confusion matrix, and an ROC curve when probabilities exist, for classification, and a residual plot and a predicted-versus-actual plot for regression, from the stored results. It shall not draw a chart of the wrong task type.	08_UT_01, 08_UT_02, 08_IT_01, 08_ST_01, 08_ST_02	5	5 of 5 pass	**Validated**
MLE-F-032	Saved evaluations and history. The system shall save every completed evaluation under the Analyst's account, list them newest first, and reopen each one with exactly the stored values.	06_IT_01, 07_IT_01, 10_UT_01, 10_UT_02	4	4 of 4 pass	**Validated**
MLE-F-033	Compare evaluations. The system shall show the metrics of two or more completed evaluations of the same task type side by side, warn and refuse when the task types differ, and never pick a winner.	09_UT_01, 09_IT_01, 09_IT_02, 09_ST_01	4	4 of 4 pass	**Validated**
MLE-F-034	PDF report. The system shall generate a PDF report for one saved evaluation or for several, built from the saved results, and shall refuse with a message when an evaluation no longer exists or none is selected.	11_UT_01, 11_UT_02, 11_UT_03, 11_IT_01, 11_ST_01	5	5 of 5 pass	**Validated**
MLE-F-040	Account list, suspend and reactivate. The system shall let an Admin see all accounts with their role and status and suspend or reactivate an account. A suspended account shall not be able to log in, and an open session shall end at once.	02_IT_03, 02_IT_04, 12_UT_01, 12_UT_02, 12_ST_01	5	5 of 5 pass	**Validated**
MLE-F-041	Role change. The system shall let an Admin change an account's role between Analyst and Admin, and the change shall apply from the account's next request.	13_UT_01, 13_ST_01	2	2 of 2 pass	**Validated**
MLE-F-042	Audit log. The system shall record every successful status change and role change with who made it, which account it affected, the old and new value and the time, and shall show the log to Admins. Refused changes shall not be logged as changes.	13_IT_03	1	1 of 1 pass	**Validated**

Req.	Requirement (summary)	Test cases	Cases	Result	Verdict
MLE-F-043	Protected accounts. The system shall refuse to let an Admin suspend or change the role of their own account, and shall refuse to suspend or demote the last active Admin.	12_IT_01, 12_IT_02, 13_IT_01, 13_IT_02	4	4 of 4 pass	**Validated**
MLE-NF-001	Performance (performance). For a dataset of up to 50,000 rows and a model file of up to 100 MB, the evaluation result shall be returned within 10 seconds for a single user.	NF_PF_01, NF_PF_02, NF_PF_03	3	3 of 3 pass	**Validated**
MLE-NF-002	Data isolation (security). Datasets, models and evaluations of one account shall not be visible or reachable by any other account, including by guessing a URL or an ID.	SEC_08, SEC_09, SEC_10, SEC_11, SEC_12	5	5 of 5 pass	**Validated**
MLE-NF-003	Clear error reporting (reliability). Invalid files, incompatible inputs and errors during evaluation shall always be reported with a specific reason; the system shall not fail silently or crash the session.	NF_ER_01, NF_ER_02, NF_ER_03, NF_ER_04	4	4 of 4 pass	**Validated**
MLE-NF-004	First-time usability (usability). A first-time user shall be able to go from uploading files to viewing results without outside instructions.	NF_US_01, NF_US_02	2	2 of 2 pass	**Validated**
MLE-NF-005	Extensibility (maintainability). The evaluation pipeline shall be organised so that a new metric, a new visualization or a new task type can be added without changing unrelated parts of the system.	NF_EX_01	1	1 of 1 pass	**Validated**
MLE-SR-001	Password hashing. Passwords shall be stored only as salted hashes and never in plain text.	SEC_01	1	1 of 1 pass	**Validated**
MLE-SR-002	Login required. Every page and API call except login, registration and static files shall require a signed-in user; anonymous page requests are sent to the login page and anonymous API calls get 401.	SEC_03	1	1 of 1 pass	**Validated**
MLE-SR-003	Role enforcement. Admin pages and the Admin API shall be refused to Analysts, and an Analyst shall not be able to change any role or status, including their own.	SEC_06, SEC_07	2	2 of 2 pass	**Validated**
MLE-SR-004	CSRF protection. Every request that changes data shall carry a valid CSRF token and shall be rejected otherwise.	SEC_05	1	1 of 1 pass	**Validated**
MLE-SR-005	Per-account isolation. Every model, dataset, evaluation, chart, comparison and report shall be scoped to its owner; requests for another account's records shall return 404.	SEC_08, SEC_09, SEC_10, SEC_11, SEC_12	5	5 of 5 pass	**Validated**
MLE-SR-006	Safe login redirect. After login the system shall redirect only to pages on its own site.	SEC_04	1	1 of 1 pass	**Validated**
MLE-SR-007	Session security. Sessions shall use HttpOnly, SameSite=Lax cookies, expire after 8 hours, and end immediately when the account is suspended.	02_IT_04	1	1 of 1 pass	**Validated**

Req.	Requirement (summary)	Test cases	Cases	Result	Verdict
MLE-SR-008	Upload limits and storage. Uploads shall be limited to the allowed file types and sizes, stored under generated names, and a rejected upload shall leave nothing stored.	03_UT_01, 03_UT_04, 04_UT_01, 04_UT_03, NF_ER_04	5	5 of 5 pass	**Validated**
MLE-SR-009	TLS 1.2+ when deployed. When the system is deployed beyond a single trusted computer, all network connections shall use TLS 1.2 or later.		0	0 of 0 pass	**NOT validated**

5. Evidence for the main end-to-end flows

These system-level cases walk through complete user journeys and are the strongest evidence that the core use cases work together:

Case	Journey	Result
06_ST_01	Upload dataset and model, validate, choose classification, evaluate, view metrics and charts (end to end)	Pass
06_ST_02	Same journey for regression; several models evaluated on one dataset	Pass
09_IT_02	Compare evaluations of the same task side by side	Pass
11_UT_01 / 11_UT_02	Download a PDF report for one and for several evaluations	Pass
13_UT_01 / 12_UT_02	Admin changes a role and suspends / reactivates an account; actions appear in the audit log (13_IT_03)	Pass
SEC_09 / SEC_10	A second Analyst cannot reach the first one's files, evaluations, charts or reports by ID	Pass

6. Limitations and observations

Validation is only as strong as the tests behind it. These points are stated openly so the results are not over-read:

MLE-NF-001. Measured on one Windows laptop (single user). Both 50,000-row runs of the suite and five runs with a 95.8 MB random-forest model (1.59 to 1.67 s) were far below 10 s. Only one kind of large model was tested.

MLE-NF-004. One classmate who was not on the team and had not seen the app finished the whole flow unaided in about 2-3 minutes with no hesitation. A single tester is a small sample.

MLE-NF-005. Checked by adding a chart type through the chart module only. Adding a new metric or task type was not exercised in a test; the design keeps them in separate modules.

MLE-SR-007. The HttpOnly and SameSite cookie flags and the 8-hour lifetime are set in app.py and were checked by reading the code; the automated test covers only that a suspension ends the session at once.

MLE-SR-009. TLS is a deployment matter. The built-in Flask server uses plain HTTP, and no test of HTTPS was run. This requirement is therefore not validated here.

MLE-F-021. Metric values are compared with hand calculations on small fixed datasets (the same applies to MLE-F-022).

Evolution since the first design. During Sprint 3 the storage moved from JSON files to SQLite and the interface became a multi-page application with login and roles. These changes support the same requirements; the test plan was updated to match (BL-12).

Known gaps outside the SRS. There is no password-reset function, and files uploaded before the move to SQLite must be uploaded again. Neither is a requirement of the SRS.

7. Conclusion

All 19 functional requirements, all 5 non-functional requirements and 8 of 9 security requirements are validated: 86 of 86 test cases passed and no defects were found. Not validated here: MLE-SR-009 (a deployment check, see section 6). The system meets the acceptance criteria of the SRS, with the limitations in section 6.